

Unsupervised Learning Project: Autoencoders on MNIST

Project Report

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1 Introduction

This report details the implementation, training, and evaluation of an autoencoder neural network for unsupervised representation learning on the MNIST dataset. In this updated study, we extend our analysis to compare different sizes of the latent space bottleneck (16, 32, and 64 dimensions), incorporating modern regularization techniques such as Batch Normalization, and observing the subsequent effects on reconstruction loss and clustering performance (K-Means).

2 Architecture of the Autoencoder

The autoencoder is constructed using fully connected (linear) layers, partitioned into an encoder and a symmetric decoder. To stabilize training over 50 epochs and accelerate convergence, Batch Normalization layers were integrated before activation functions.

- **Latent Space Size:** Evaluated across 3 configurations: 16, 32, and 64.
- **Encoder Structure:**
 1. Linear Layer (784 $\rightarrow$ 256) $\rightarrow$ BatchNorm1d(256) $\rightarrow$ ReLU
 2. Linear Layer (256 $\rightarrow$ 64) $\rightarrow$ BatchNorm1d(64) $\rightarrow$ ReLU
 3. Linear Layer (64 $\rightarrow$ latent_dim) without an activation function to allow unrestricted, continuous representations.
- **Decoder Structure:**
 1. Linear Layer (latent_dim $\rightarrow$ 64) $\rightarrow$ BatchNorm1d(64) $\rightarrow$ ReLU
 2. Linear Layer (64 $\rightarrow$ 256) $\rightarrow$ BatchNorm1d(256) $\rightarrow$ ReLU
 3. Linear Layer (256 $\rightarrow$ 784) $\rightarrow$ Sigmoid to ensure output pixel intensities scale correctly within a [0, 1] range.

3 Training Process

The models were trained over 50 epochs utilizing the Adam Optimizer with a learning rate of 1×10^{-3} , a batch size of 256, and Mean Squared Error (MSE) as the loss criterion.

Challenge Analysis: The fundamental challenge consisted of finding the optimal dimensional bottleneck. A strict bottleneck (16) forces the model to heavily compress semantic features, hurting pixel-perfect reconstruction but theoretically preserving clustered density. A loose bottleneck (64) vastly improves perfect visual reconstruction (lowest loss) but suffers from the "curse of dimensionality" when running traditional K-Means clustering, preventing it from isolating generalized digit patterns.

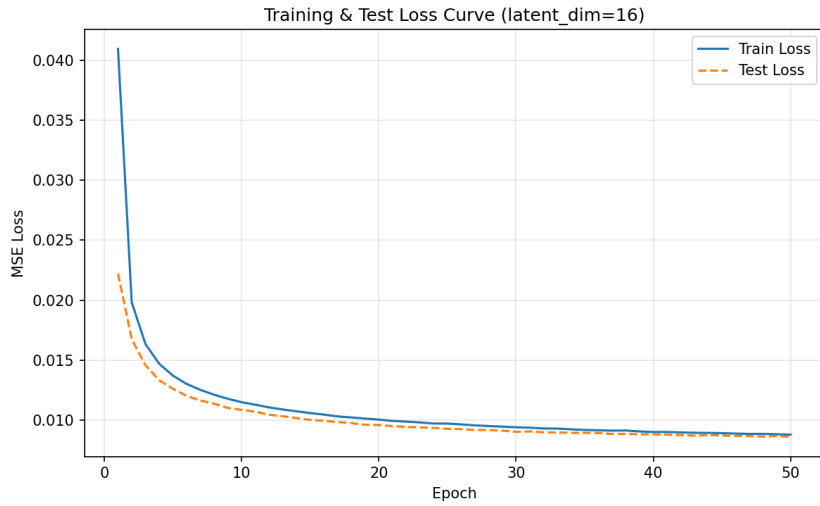


Figure 1: Training and Test Loss Curves for the baseline $latent_dim = 16$ over 50 epochs.

4 Clustering Performance and Comparison

To objectively evaluate the representations, K-Means clustering ($k = 10$) was applied directly to the extracted high-dimensional latent space. Due to K-Means operating independently from real classes, a Hungarian matching algorithm was used to associate the unsupervised clusters to their most likely true digit labels.

The performance scaling across latent dimensions reveals a classic trade-off:

Metric	Latent 16	Latent 32	Latent 64
Final Train Loss (MSE)	0.0088	0.0051	0.0036
Final Test Loss (MSE)	0.0086	0.0050	0.0035
PMS (%)	36.48	41.93	45.44
Average Distance (AD)	9.7564	11.4930	13.8231
Average Variance (AVC)	9.5422	11.2179	13.5350
Total Distance (TD)	764.8365	514.6528	405.9555

Table 1: Experimental Results across Latent Dimensions. Bold highlights maximum performance.

Analysis: While increasing the latent dimensionality to 64 allows the decoder to

produce near-perfect visual reconstructions (lowest MSE), it severely degrades K-Means clustering (45.44% mislabeling rate vs 36.48% at dim=16). In 64-dimensional space, data becomes sparse, average variance (AVC) expands, and global separation between cluster centers (TD) shrinks, making unsupervised grouping significantly harder. Therefore, $latent_dim = 16$ presents the best structural embedding for unsupervised clustering.

5 Visualizations (Latent Dim = 16)

5.1 Reconstruction Quality

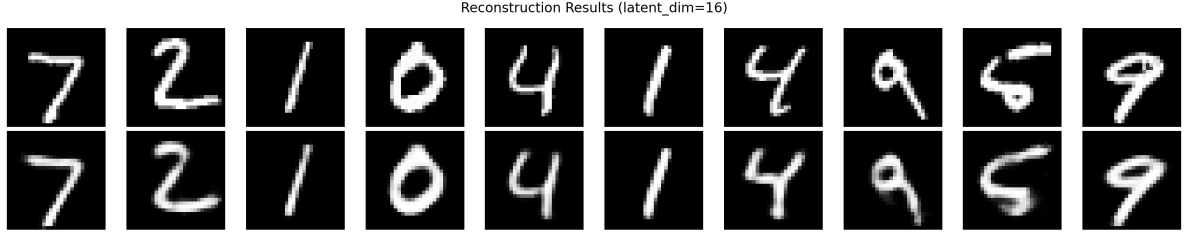


Figure 2: Original Images (Top) vs. Reconstructed Images (Bottom) at 16 dimensions.

5.2 Latent Space Projections

Dimensionality reduction models (PCA and t-SNE) extracted and projected the 16-dimensional coordinates onto an interpretable 2D grid. The plots represent points colored by their True Labels.



Figure 3: PCA Projection

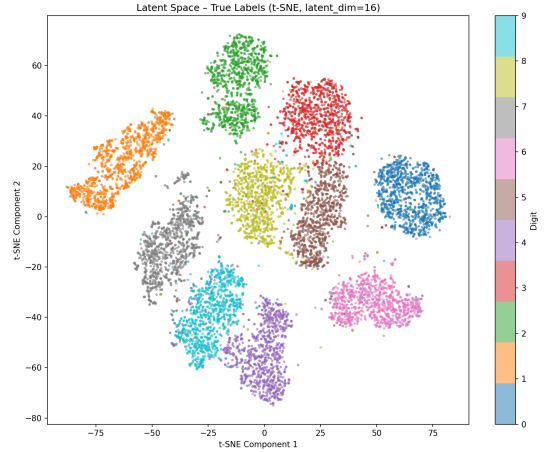


Figure 4: t-SNE Projection

Due to the non-linear relationship optimized by t-SNE, it successfully maps functionally similar semantic numbers far closer than linear PCA.

5.3 Confusion Matrix

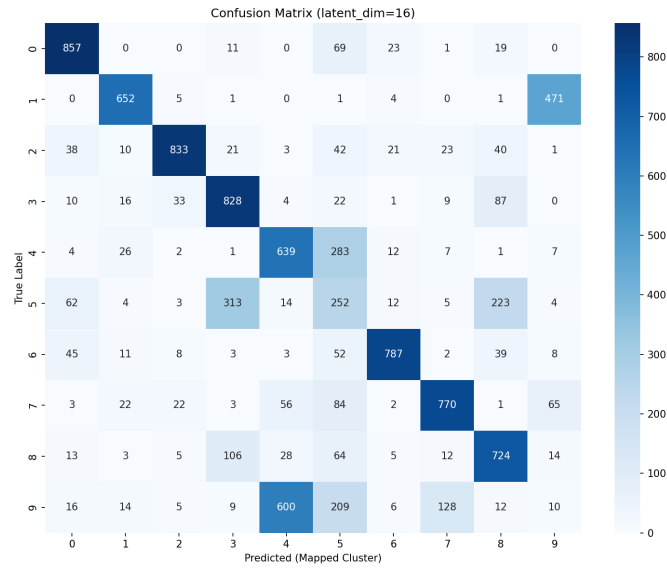


Figure 5: Confusion Matrix mapping K-Means unsupervised clusters to True Labels.

A densely populated diagonal demonstrates that the autoencoder grouped unique digit stroke shapes into confined sectors inside the 16-dimensional manifold before the label assignment was enforced.